

Hybrid Collatz Sequence–Based Regularization for Improving Stability and Generalization in Deep Neural Networks

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I. BACKGROUND

Deep neural networks (DNNs) achieve high performance through a delicate interaction between optimization dynamics, architectural design, initialization strategies, and regularization mechanisms. Among these, weight initialization and regularization play a particularly critical role in determining gradient flow, convergence behavior, and generalization performance.

Initialization and Training Dynamics

Weight initialization determines the starting point of optimization and strongly influences early gradient propagation, symmetry breaking, and convergence speed. Classical randomized schemes such as Xavier initialization [1] and He initialization [2] are designed to preserve signal variance across layers, thereby preventing vanishing or exploding gradients. These methods are highly effective and remain the default choice in most modern architectures.

However, randomized initialization introduces inherent run-to-run variability. While such variability can be beneficial by implicitly promoting exploration of the loss landscape, it also reduces reproducibility and can lead to inconsistent convergence behavior across training runs, particularly in smaller datasets or shallow networks. This has motivated recent interest in deterministic and structured initialization strategies that aim to stabilize early training while preserving sufficient expressiveness.

Recent work on deterministic initialization demonstrates that carefully designed structure can improve optimization stability without sacrificing performance. In particular, the Collatz Sequence Initialization method [3] introduces number-theoretic structure into weight matrices and empirically shows reduced gradient variance and faster convergence compared to purely random initialization. Similar stability-oriented approaches include Sinusoidal Initialization [4], which uses periodic patterns to control spectral properties of weights, and IDInit [5], which approximates identity mappings to ease signal propagation in deep networks.

Importantly, these approaches operate *only at initialization* and do not actively influence the optimization trajectory once training begins.

Regularization and the Role of Noise

Regularization techniques are designed to improve generalization by controlling model complexity and shaping the optimization trajectory. Classical methods such as dropout [6],

data augmentation, and minibatch stochastic gradient descent (SGD) rely on stochasticity or noise injection. Extensive theoretical and empirical evidence shows that such noise plays a constructive role: it encourages exploration of the loss landscape, biases optimization toward flatter minima, and improves generalization [7], [8].

Therefore, stochastic regularization should not be viewed as inherently destabilizing. Rather, instability typically arises when noise interacts poorly with learning rates, architectural choices, or poorly conditioned parameter spaces. In such cases, excessive stochasticity may slow convergence, amplify gradient variance, or obscure reproducibility.

This observation motivates the search for complementary mechanisms that can *guide* optimization trajectories without eliminating the beneficial effects of stochasticity.

From Deterministic Initialization to Deterministic Regularization

While deterministic initialization constrains the starting point of training, it does not prevent weights from drifting into highly irregular or noisy configurations as optimization progresses. This limitation suggests a conceptual gap between deterministic structure at initialization and the largely unconstrained dynamics of training-time regularization.

Recent work on Deterministic Feature Regularization (DFReg) [9] provides early evidence that deterministic constraints can be applied during training to stabilize internal representations. DFReg demonstrates that deterministic priors can coexist with stochastic optimization and may reduce run-to-run variance while preserving generalization.

However, existing deterministic regularizers operate primarily at the level of feature representations rather than directly constraining weight evolution. Moreover, no prior work has explored number-theoretic or sequence-based deterministic structure as a training-time regularization mechanism.

Motivation for Collatz-Based Regularization

The Collatz sequence exhibits simple deterministic rules yet generates complex, nontrivial patterns. Its recent successful application to weight initialization suggests that such structure can bias optimization toward smoother and more stable trajectories. This raises a natural and currently unexplored question: *can Collatz-based structure serve as a soft, training-time regularizer rather than merely an initialization scheme?*

Rather than replacing stochastic regularization, a Collatz Sequence-Based (CSB) regularizer can be viewed as introducing a deterministic inductive bias that gently constrains weight evolution while allowing stochastic methods such as dropout and SGD to continue promoting exploration. In this sense, CSB regularization is not intended to suppress noise, but to complement it by providing structural guidance.

This perspective aligns with modern views of optimization as a balance between exploration (stochasticity) and control (structure), and motivates the investigation of hybrid deterministic–stochastic regularization frameworks for improving stability, reproducibility, and generalization in deep neural networks.

II. MOTIVATION FOR THE STUDY

Recent advances in deterministic initialization methods have demonstrated that structured, non-random weight patterns can significantly improve early optimization stability and reduce run-to-run variance in deep neural networks. In particular, Collatz Sequence Initialization [3] shows that number-theoretic structure can bias optimization toward smoother gradient dynamics and faster convergence without degrading predictive performance. However, this benefit is limited to the initial phase of training and does not persist once standard stochastic optimization takes over.

In contrast, regularization techniques such as dropout, L2 weight decay, and stochastic gradient descent (SGD) operate throughout training and are known to improve generalization by injecting noise and encouraging exploration of the loss landscape. Modern optimization theory suggests that such stochasticity biases learning toward flatter minima, which are associated with better generalization. Nevertheless, excessive or poorly controlled noise can increase gradient variance, reduce reproducibility, and slow convergence, particularly in small or medium-scale models.

These observations suggest a conceptual imbalance in current training pipelines: while deterministic structure is carefully designed at initialization, training-time regularization is dominated by stochastic mechanisms with limited structural guidance. As a result, the optimization trajectory may drift away from the stable regions encouraged at initialization, potentially diminishing the early benefits of deterministic design.

Recent work on deterministic regularization, such as Deterministic Feature Regularization (DFReg) [9], provides early evidence that deterministic constraints can be applied during training without eliminating the beneficial effects of stochastic optimization. However, existing approaches focus primarily on stabilizing feature representations rather than directly shaping weight evolution, and they do not explore number-theoretic or sequence-based structure.

This gap motivates the present study. Given the empirical success of Collatz-based structure at initialization, it is natural to ask whether similar structure can be extended into a training-time regularization mechanism. Rather than replacing stochastic regularizers, a Collatz Sequence-Based

(CSB) regularizer may act as a soft deterministic inductive bias that guides weight evolution while preserving stochastic exploration through dropout and SGD.

By investigating a hybrid deterministic–stochastic regularization framework, this study aims to determine whether Collatz-based structure can (i) stabilize optimization trajectories, (ii) reduce run-to-run variability, and (iii) improve generalization when combined with standard regularization techniques. Addressing this question contributes to the emerging research direction that seeks to balance deterministic control and stochastic exploration in deep neural network training.

III. STATEMENT OF THE PROBLEM

Deep neural network training relies on a combination of initialization strategies and regularization mechanisms to achieve stable optimization and strong generalization. While recent deterministic initialization methods, such as Collatz Sequence Initialization [3], have shown clear benefits in reducing early-stage gradient variance and improving convergence behavior, their influence is limited to the starting point of training and does not actively constrain weight evolution thereafter.

In contrast, training-time regularization is dominated by stochastic techniques, including dropout, and minibatch stochastic gradient descent. These methods are well-known to provide strong implicit regularization by injecting noise that encourages exploration of the loss landscape and biases optimization toward flatter minima. However, their stochastic nature also introduces run-to-run variability and can obscure the relationship between initialization structure and final learned representations.

As a result, there exists a conceptual and practical gap in current deep learning training pipelines: deterministic structure is carefully engineered at initialization, but largely abandoned during training, leaving optimization trajectories guided almost exclusively by stochastic forces. It remains unclear whether the stability benefits introduced by deterministic initialization can be preserved or extended through training-time regularization.

Recent work on deterministic regularization, such as Deterministic Feature Regularization (DFReg) [9], suggests that deterministic constraints can coexist with stochastic optimization and may reduce variance across runs. Nevertheless, existing approaches focus on feature-level stabilization and do not address the role of deterministic structure in guiding weight evolution. Moreover, no prior work has investigated number-theoretic or sequence-based deterministic constraints—such as those derived from the Collatz sequence—as explicit regularization mechanisms during training.

Therefore, the central problem addressed in this study is the lack of understanding of whether Collatz Sequence-Based deterministic structure can function as an effective training-time regularizer, and how its integration with standard stochastic regularization methods affects optimization stability, generalization performance, reproducibility, and computational cost in deep neural networks.

IV. RESEARCH QUESTIONS

This study seeks to investigate the role of Collatz Sequence–Based (CSB) deterministic structuring as a training-time regularization mechanism in deep neural networks, both independently and in combination with standard stochastic regularization techniques. The research is guided by the following questions:

- 1) Does applying Collatz Sequence–Based deterministic structuring during training improve gradient stability and convergence when combined with traditional regularization techniques?
- 2) To what extent does a hybrid deterministic–stochastic regularization framework reduce overfitting and improve generalization across neural architectures and benchmark datasets?
- 3) What computational overhead, trade-offs, and limitations arise from integrating CSB-based deterministic regularization with existing stochastic methods?

V. OBJECTIVES OF THE STUDY

General Objective

The general objective of this study is to design, implement, and empirically evaluate a Hybrid Collatz Sequence–Based (CSB) Regularization framework that integrates deterministic weight structuring with standard stochastic regularization techniques to improve training stability, generalization performance, and reproducibility in deep neural networks.

Specific Objectives

The specific objectives of this study are to:

- Formulate a Collatz Sequence–Based deterministic regularization term that can be applied during training to constrain weight evolution in a soft and flexible manner.
- Integrate the proposed CSB regularization with commonly used regularization methods, including L2 weight decay and dropout, within standard deep learning training pipelines.
- Empirically analyze the impact of CSB regularization on optimization stability by examining gradient norms, loss trajectories, weight norms, and run-to-run variance across multiple random seeds.
- Evaluate the effect of the hybrid deterministic–stochastic regularization framework on generalization performance and generalization gap across multiple architectures and benchmark datasets.
- Conduct ablation studies to assess the sensitivity of CSB regularization to hyperparameters and to identify conditions under which it provides benefits or introduces performance trade-offs.
- Measure and compare the computational overhead introduced by CSB regularization in terms of training time and memory usage relative to standard regularization baselines.

VI. EXPECTED CONTRIBUTIONS

This study is expected to make the following contributions to the field of deep learning optimization and regularization:

- **A novel training-time deterministic regularization framework:** The study introduces Collatz Sequence–Based (CSB) deterministic structuring as an explicit regularization mechanism applied during training, extending prior work that employs Collatz-based structure only at initialization.
- **Empirical evidence on hybrid deterministic–stochastic regularization:** Through systematic experimentation, the study provides empirical insights into how CSB regularization interacts with standard methods such as L2 weight decay and dropout, clarifying when hybrid regularization improves stability and generalization and when it introduces trade-offs.
- **Analysis of optimization stability and reproducibility:** The work offers a detailed analysis of gradient dynamics, loss trajectories, weight evolution, and run-to-run variance, contributing to a better understanding of how deterministic structure influences optimization behavior under stochastic training.
- **Assessment of computational overhead and feasibility:** The study quantifies the computational and memory costs associated with CSB regularization, providing practical guidance on its feasibility and scalability for common neural network architectures.
- **Contribution to the emerging theory of structured regularization:** By grounding CSB regularization in optimization theory and flat-minima perspectives, this work contributes to the broader research direction that investigates how deterministic inductive biases can complement stochastic implicit regularization in deep neural networks.

VII. SCOPE AND DELIMITATIONS

Scope

The scope of this study is limited to the empirical evaluation of Collatz Sequence–Based (CSB) deterministic regularization within standard supervised deep learning settings. In particular, the study focuses on:

- **Neural architectures:** Multilayer perceptrons (MLPs) and convolutional neural networks (CNNs) with ReLU-based activations.
- **Regularization techniques:** Collatz Sequence–Based deterministic regularization, L2 weight decay, and dropout, applied individually and in hybrid configurations.
- **Learning paradigm:** Supervised classification tasks trained using stochastic gradient-based optimization.
- **Datasets:** Standard benchmark image datasets, namely MNIST, Fashion-MNIST, and CIFAR-10.
- **Evaluation criteria:** Generalization performance (accuracy and generalization gap), optimization stability (loss trajectories, gradient norms, and run-to-run variance), and computational cost (training time and memory usage).

Delimitations

To maintain focus and ensure feasibility within the scope of a Master’s-level study, the following delimitations are imposed:

- Transformer-based and large-scale architectures are excluded due to their substantially different optimization dynamics and computational requirements. Investigating CSB regularization in such models is left for future work.
- The study does not address unsupervised, self-supervised, or reinforcement learning settings, and focuses exclusively on supervised learning tasks.
- Only Collatz sequence-based deterministic structuring is considered. Other number-theoretic, chaotic, or deterministic sequences are beyond the scope of this research.
- Hyperparameter optimization is limited to controlled experimental sweeps and is not intended to provide globally optimal tuning for each model–dataset combination.
- Theoretical analysis is limited to conceptual interpretation and empirical validation; formal convergence proofs or generalization bounds are not pursued.

VIII. LITERATURE REVIEW

The performance of deep neural networks (DNNs) critically depends on both how models are initialized and how they are regularized during training. Regularization encompasses a broad range of techniques designed to improve generalization and reduce overfitting, while initialization determines the trajectory of early optimization. Recent research from 2023 to 2025 has seen significant advances in both domains, as well as efforts to bridge structural and stochastic approaches.

A. Traditional and Stochastic Regularization Techniques

Stochastic regularization methods remain central to deep learning practice. Classical techniques such as L2 weight decay, dropout, and early stopping have demonstrated strong empirical performance across tasks. Surveys of dropout approaches show that numerous variants (e.g., adaptive dropout, structured dropout) have been developed to control noise injection while moderating model complexity [10]. By randomly masking units or weights during training, dropout forces the network to develop redundant and robust representations, which generally improves generalization.

Stochastic gradient descent (SGD) itself serves as an implicit regularizer. Recent analysis reveals that SGD noise biases training toward “flat” minima of the loss landscape—a property associated with improved generalization [11]. These implicit effects arise even in the absence of explicit regularization terms, suggesting that noise and stochasticity are not merely artifacts but fundamental components of successful optimization.

Techniques like dynamic sparse training have recently been shown to act as effective implicit regularizers that improve robustness against data corruption, indicating that sparsity and noise can be synergistic [12]. Such methods challenge the view that explicit regularization is always necessary, suggesting that

optimization strategy and model dynamics may themselves serve regularization roles.

Despite their successes, stochastic regularizers can exhibit run-to-run variability, particularly in small datasets or low-noise regimes. The randomness introduced by dropout, mini-batch sampling, and data augmentation may complicate reproducibility and require careful hyperparameter tuning.

B. Deterministic and Learned Regularization

Beyond stochastic methods, there has been growing interest in deterministic forms of regularization and weight control. These include spectral regularization, explicit learned regularizers, and physics-inspired frameworks that shape weight evolution beyond simple penalty terms.

Spectral regularization techniques control the spectrum of weight matrices to influence generalization behavior. Research in spectral filtering and neural tangent kernel (NTK) analysis provides theoretical insight into how regularization affects function complexity and convergence properties [13]. These perspectives link regularization with classical ideas from inverse problems and operator theory.

Recent work on learned regularizers explores how neural networks can be trained to impose explicit structure on solutions. For imaging inverse problems, explicit learned regularizers have been shown to outperform implicit CNN priors while retaining convergence guarantees [14]. Such methods represent a departure from manually designed penalty terms, instead learning structured regularization directly from data and optimization dynamics.

Another direction is the development of deterministic regularizers applied during training. DFReg, a physics-inspired framework, imposes a global constraint on the weight distribution, encouraging smooth and well-spread weight patterns without reliance on stochastic penalties [9]. This work highlights that deterministic regularization can match or even outperform traditional stochastic regularizers, particularly in scenarios where architectural changes (e.g., batch normalization) are not feasible.

C. Deterministic Initialization and Structured Inductive Bias

While deterministic regularization during training is still emerging, deterministic initialization methods have gained traction. Collatz Sequence Initialization introduces number-theoretic structure into weight matrices, reducing early-training gradient variance and accelerating convergence [3]. Similarly, sinusoidal and identity-inspired initializations impose smooth or identity-like structure to improve signal propagation in deep layers [4], [5]. These methods show that structured initial weights can reduce optimization instability and improve reproducibility.

However, deterministic initialization alone does not persist as a regularizing force throughout training. Once stochastic optimization begins, the initial structure may dissipate. This highlights a gap between the benefits of structured starting conditions and the lack of persistent structural priors during learning.

D. Theoretical Perspectives: Optimization, Flat Minima, and Implicit Bias

A unifying theme in modern regularization research is the connection between optimization dynamics and generalization. The flat minima hypothesis suggests that minima with wide, low-curvature basins correlate with robust generalization. Noise from SGD and dropout is believed to steer training toward such regions [11]. Complementary research has emphasized that deterministic patterns can bias gradients and curvature properties, thus influencing the geometry of the learned model.

Implicit bias theory highlights that even in overparameterized models, optimization algorithms inherently prefer certain solutions, which acts as an implicit regularizer. The interaction between explicit penalties and implicit biases remains an active area of study, especially in understanding how different regularizers shape the loss landscape and model complexity.

E. Identified Research Gap

Despite extensive work on stochastic regularizers, structured initializations, and learned regularizers, there remains a lack of research on sequence-based deterministic regularization applied directly during training. In particular, the extension of number-theoretic structures like Collatz sequences from initialization into persistent regularizers has not been explored.

Moreover, little is known about how such deterministic structures interact with classical stochastic methods like dropout and weight decay, or how they influence optimization trajectories, stability, and generalization jointly. Addressing this gap motivates the present study of hybrid deterministic–stochastic regularization frameworks.

F. Theoretical Rationale

Modern deep learning optimization can be understood as a dynamic interplay between deterministic forces, which guide descent along the loss gradient, and stochastic forces, which introduce noise that enables exploration of the loss landscape. Stochastic gradient descent (SGD), dropout, and data augmentation are now widely recognized as sources of implicit regularization that bias optimization toward flatter minima and low-complexity solutions. Such minima are empirically and theoretically associated with improved generalization.

At the same time, excessive or poorly controlled stochasticity can increase gradient variance, slow convergence, and reduce reproducibility across training runs. From this perspective, stability is not achieved by eliminating noise, but by shaping how noise interacts with the underlying optimization trajectory.

Deterministic initialization methods provide an instructive example of this balance. By imposing structured priors on weights at the start of training, methods such as Collatz Sequence Initialization [3] reduce early-stage instability while still allowing SGD-induced noise to explore the parameter space. However, once training begins, these deterministic priors rapidly dissipate, and subsequent weight evolution is

governed almost entirely by stochastic regularization mechanisms.

The central theoretical hypothesis of this work is that extending deterministic structure into training-time regularization can provide a persistent inductive bias that guides optimization without suppressing beneficial stochastic exploration. Specifically, a Collatz Sequence–Based (CSB) regularizer can be viewed as a soft constraint that penalizes deviation from a structured reference trajectory in weight space. Unlike hard constraints, such a penalty does not restrict the hypothesis class, but instead biases optimization toward smoother and more stable regions of the loss landscape.

From the perspective of flat-minima theory, the CSB regularizer may reduce sensitivity to parameter perturbations by discouraging sharp curvature directions, while stochastic methods continue to encourage basin exploration. Similarly, in terms of implicit bias, CSB regularization introduces an explicit, deterministic preference that interacts with the implicit bias of SGD, potentially yielding solutions that combine stability, reproducibility, and strong generalization.

This theoretical framing positions CSB regularization not as a replacement for stochastic regularization, but as a complementary mechanism that balances deterministic control and stochastic exploration. The empirical component of this study is therefore designed to test whether this balance can be achieved in practice and under what conditions it yields measurable benefits.

IX. METHODOLOGY

This study adopts an experimental and comparative research design to investigate the effectiveness of Collatz Sequence–Based (CSB) deterministic regularization as a training-time regularization mechanism in deep neural networks. The methodology is designed to evaluate optimization stability, generalization performance, and computational feasibility under both deterministic and stochastic regularization settings. The overall workflow consists of five main stages: (i) design of the CSB regularizer, (ii) integration with stochastic regularization techniques, (iii) experimental setup and training protocol, (iv) evaluation metrics, and (v) statistical and ablation analysis.

A. Design of the Collatz Sequence–Based Regularizer

The proposed CSB regularization method extends the idea of Collatz Sequence Initialization by introducing deterministic structure as a persistent, training-time constraint on network weights. The Collatz transformation is defined as

$$T(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ 3n + 1, & \text{if } n \text{ is odd.} \end{cases}$$

Repeated application of this transformation generates a Collatz sequence $\mathbf{s} = (s_1, s_2, \dots, s_K)$, which is truncated or tiled to match the dimensionality of a given weight matrix.

For each selected layer with weight matrix $W \in \mathbb{R}^{m \times n}$, a structured reference matrix W_{CSB} is constructed by reshaping the normalized Collatz sequence into the same shape as W .

To ensure numerical stability, the sequence is standardized to zero mean and unit variance before reshaping.

The CSB regularization term is defined as a soft penalty on deviation from the structured reference:

$$\mathcal{L}_{\text{CSB}}(W) = \lambda_{\text{CSB}} \|W - W_{\text{CSB}}\|_F^2,$$

where λ_{CSB} controls the strength of the deterministic constraint and $\|\cdot\|_F$ denotes the Frobenius norm. This formulation biases weight evolution toward structured trajectories while preserving flexibility for stochastic optimization.

B. Integration with Stochastic Regularization

To study hybrid deterministic–stochastic regularization, the CSB penalty is combined with standard regularization techniques commonly used in deep learning. Let $\mathcal{L}_{\text{task}}$ denote the task-specific loss function (cross-entropy for classification). The total training objective is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{L2}} \|W\|_2^2 + \sum_{l \in \mathcal{S}} \mathcal{L}_{\text{CSB}}(W^{(l)}),$$

where λ_{L2} is the weight decay coefficient and $\mathcal{S}$ denotes the set of layers to which CSB regularization is applied.

Dropout is applied to hidden layers with predefined drop probabilities, and batch normalization is used where appropriate to stabilize activation distributions. Multiple training configurations are evaluated, including: (i) baseline stochastic regularization, (ii) CSB-only regularization, and (iii) hybrid CSB combined with L2 weight decay and dropout. This design enables direct comparison of deterministic, stochastic, and hybrid approaches.

C. Experimental Setup

1) *Datasets*: Experiments are conducted on widely used supervised image classification benchmarks: MNIST, Fashion-MNIST, and CIFAR-10. These datasets provide increasing levels of complexity and allow controlled evaluation of generalization and stability across domains. Standard training, validation, and test splits are used, with a subset of training data reserved for validation and early stopping.

2) *Neural Architectures*: The study focuses on multi-layer perceptrons (MLPs) and convolutional neural networks (CNNs). MLPs consist of 3–4 fully connected hidden layers with ReLU activations, while CNNs follow compact architectures such as LeNet-style or small residual models. This selection balances representational power with computational feasibility.

3) *Training Protocol*: All models are trained under controlled and comparable conditions. Stochastic Gradient Descent (SGD) with momentum or Adam is used as the optimizer, with learning rates selected via preliminary tuning and optionally decayed using step or cosine schedules. Batch sizes range from 64 to 256, depending on dataset and architecture. Training is performed for a fixed maximum number of epochs, with early stopping based on validation performance.

To isolate the effect of regularization, all experiments use a standard random initialization scheme (He or Xavier). Each

configuration is repeated across multiple random seeds to measure run-to-run variability and reproducibility.

D. Evaluation Metrics

Evaluation focuses on three complementary dimensions: generalization, optimization stability, and computational cost.

1) *Generalization Performance*: Generalization is assessed using training, validation, and test accuracy, as well as the generalization gap between training and test performance. Loss curves are analyzed to inspect convergence behavior.

2) *Optimization Stability*: Optimization stability is evaluated through gradient norm tracking, loss oscillation analysis, and monitoring of layer-wise weight norms. Reduced variance across runs is interpreted as improved stability and reproducibility.

3) *Computational Cost*: Computational feasibility is assessed by measuring wall-clock training time per epoch, total training time, and approximate memory usage. Additional overhead introduced by CSB regularization is explicitly quantified.

E. Statistical and Ablation Analysis

To ensure robustness, results are aggregated across multiple runs, reporting mean and standard deviation for all metrics. Statistical significance is evaluated using paired tests or analysis of variance (ANOVA) at a standard significance level.

Ablation studies are conducted to isolate the contribution of CSB regularization by varying its strength, layer coverage, and interaction with stochastic regularizers. These analyses identify regimes in which CSB provides measurable benefits and clarify potential trade-offs.

Overall, this methodology provides a systematic framework for evaluating Collatz Sequence–Based deterministic regularization within modern deep learning training pipelines.

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